THE UNIVERSITY OF MELBOURNE

| Student number |
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Semester 1 Assignment 2, 2025

School of Mathematics and Statistics

# MAST20009 Vector Calculus

Submission deadline: 5pm on Monday April 14

This assignment consists of 11 pages (including this page) with 4 questions and 45 total marks

## Instructions to Students

- This assignment is worth 5% of your total MAST20009 mark.
- If you have a printer, print the assignment one-sided.

*Writing*

- Assignments must be neatly written in blue or black pen or may be typeset using LaTeX.
- Full working must be shown in your solutions.
- Marks will be deducted for incomplete working, insufficient justification of steps, incorrect mathematical notation and for messy presentation of solutions.
- If you are writing answers on the assignment and you need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

*Scanning and Submitting*

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned assignment as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary.

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**Question 1 (11 marks)**

(a) Let $f(x, y) = x^3 + y^2 - 5xy + 2x$. Find its critical points.

We have

$$f_x(x, y) = 3x^2 - 5y + 2$$
$$f_y(x, y) = 2y - 5x$$

Setting $f_x(x, y) = f_y(x, y) = 0$, we get

$$3x^2 - 5y + 2 = 0 \quad (1)$$
$$2y - 5x = 0 \quad (2)$$

Taking $2 \times (1) + 5 \times (2)$ yields

$$4 + 6x^2 - 25x = 0$$

and hence $x = \frac{1}{6}$ or $x = 4$. Substituting back into (2) we get the two critical points $(\frac{1}{6}, \frac{5}{12})$ and $(4, 10)$.

(b) Determine the types of the critical points.

The second-order partial derivatives are

$$f_{xx}(x,y) = 6x, \quad f_{xy}(x,y) = -5, \quad f_{yy}(x,y) = 2$$

so the Hessian determinant is

$$\det H(x,y) = 12x - 25.$$

Then

$$\det H(\tfrac{1}{6}, \tfrac{5}{12}) = -23$$
$$\det H(4,10) = 23.$$

So $(\frac{1}{6}, \frac{5}{12})$ is a saddle point.

For $(4,10)$, note that $f_{xx}(4,10) = 24 > 0$, so this point is a local minimum.

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**Question 2 (10 marks)**

Let $f(x, y, z) = (x + z)e^{2y}$. Use Lagrange multipliers to find the extrema of $f$, subject to the constraint $x^2 + y^2 + z^2 = \frac{3}{2}$. Also classify the extrema.

Set $g(x, y, z) = x^2 + y^2 + z^2 - \frac{3}{2}$. Then $\nabla g = (2x, 2y, 2z)$ which clearly cannot be $(0, 0, 0)$ if $g(x, y, z) = 0$. So we should look for solutions to

$$\nabla f = \lambda \nabla g,$$

that is

$$\begin{aligned} x^2 + y^2 + z^2 &= \tfrac{3}{2} && (3)\\ e^{2y} &= 2\lambda x && (4)\\ 2(x + z)e^{2y} &= 2\lambda y && (5)\\ e^{2y} &= 2\lambda z && (6) \end{aligned}$$

We have $e^{2y} > 0$ and hence $\lambda \neq 0$. From (4) and (6) we conclude $z = x$.

Substituting into (5) and then combining with (4) gives $y = 4x^2$.

Now substituting into (3) we get

$$16x^4 + 2x^2 - \tfrac{3}{2} = \tfrac{1}{2}(4x^2 - 1)(8x^2 + 3) = 0$$

The only real solutions to this are $x = \pm\frac{1}{2}$. Working backwards, we get two solutions to our system of equations:

$$(x, y, z) = (\tfrac{1}{2}, 1, \tfrac{1}{2}) \quad \text{and} \quad (x, y, z) = (-\tfrac{1}{2}, 1, -\tfrac{1}{2})$$

The region in $\mathbb{R}^3$ satisfying the constraint is closed and bounded, so a maximum and a minimum must exist. The two points above are the only candidates.

Evaluating the function at the points, we find

$$\begin{aligned} f(\tfrac{1}{2}, 1, \tfrac{1}{2}) &= e^2\\ f(-\tfrac{1}{2}, 1, -\tfrac{1}{2}) &= -e^2. \end{aligned}$$

So there is a maximum at $(\frac{1}{2}, 1, \frac{1}{2})$ and a minimum at $(-\frac{1}{2}, 1, -\frac{1}{2})$.

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**Question 3 (10 marks)**

Consider a particle travelling along a path given by

$$\mathbf{c}(t) = (3t, 1, t^3), \quad t \geq 0.$$

(a) Find the velocity and speed of the particle.

We have

$$\text{velocity} = \frac{d\mathbf{c}}{dt} = (3, 0, 3t^2)$$
$$\text{speed} = \left|\frac{d\mathbf{c}}{dt}\right| = 3\sqrt{1+t^4}.$$

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(b) Find the unit tangent vector $\mathbf{T}(t)$, the principal normal vector $\mathbf{N}(t)$, the binormal vector $\mathbf{B}(t)$, the curvature $\kappa(t)$, and the torsion $\tau(t)$ at time $t$.

$$\mathbf{T}(t) = \frac{\mathbf{c}'(t)}{|\mathbf{c}'(t)|} = \frac{1}{\sqrt{1+t^4}}(1, 0, t^2)$$

Then

$$\begin{aligned}\mathbf{T}'(t) &= \frac{-2t^3}{(1+t^4)^{3/2}}(1, 0, t^2) + \frac{1}{\sqrt{1+t^4}}(0, 0, 2t) \\ &= \frac{1}{(1+t^4)^{3/2}}(-2t^3, 0, 2t)\end{aligned}$$

with

$$|\mathbf{T}'(t)| = \frac{2t}{1+t^4}$$

so

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|} = \frac{1}{\sqrt{1+t^4}}(-t^2, 0, 1)$$

Next

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) = \frac{1}{1+t^4}(0, -(1+t^4), 0) = (0, -1, 0)$$

For curvature,

$$\kappa(t) = \frac{|\mathbf{T}'(t)|}{|\mathbf{c}'(t)|} = \frac{2t}{1+t^4} \cdot \frac{1}{3\sqrt{1+t^4}} = \frac{2t}{3(1+t^4)^{3/2}}$$

Finally we clearly have

$$\mathbf{B}'(t) = \mathbf{0}$$

implying that $\tau(t) = 0$.

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**Question 4 (14 marks)**

Consider the vector field

$$\mathbf{F} = (x^2yz + xz)\mathbf{i} + (ax^3z + y + z)\mathbf{j} + (bx^3y + cx^2 + dy)\mathbf{k}$$

where $a, b, c, d \in \mathbb{R}$ are constants.

(a) Find the values of $a, b, c, d$ for which $\mathbf{F}$ is a gradient field.

*Note: You should double, triple, quadruple check your answers because marks for (b) will not be consequential on any mistakes in (a).*

$\mathbf{F}$ is a gradient field iff $\nabla \times \mathbf{F} = \mathbf{0}$. Now

$$\begin{aligned}\nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ x^2yz + xz & ax^3z + y + z & bx^3y + cx^2 + dy \end{vmatrix} \\ &= ((b-a)x^3 + d - 1)\,\mathbf{i} - ((3b-1)x^2y + (2c-1)x)\,\mathbf{j} \\ &\qquad + ((3a-1)x^2z)\,\mathbf{k}\end{aligned}$$

This is $(0, 0, 0)$ iff $a = b = \frac{1}{3}$, $c = \frac{1}{2}$ and $d = 1$.

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(b) Find all scalar potentials for the gradient field $\mathbf{F}$ you found in (a). Explain your reasoning.

From (a) we have $\mathbf{F} = (x^2yz + xz, \frac{1}{3}x^3z + y + z, \frac{1}{3}x^3y + \frac{1}{2}x^2 + y)$.

A scalar function $\phi(x, y, z)$ is a scalar potential for $\mathbf{F}$ if $\nabla\phi = \mathbf{F}$. So we require

$$\phi_x = x^2yz + xz \tag{7}$$

$$\phi_y = \frac{1}{3}x^3z + y + z \tag{8}$$

$$\phi_z = \frac{1}{3}x^3y + \frac{1}{2}x^2 + y \tag{9}$$

Integrating both sides of (7) with respect to $x$ gives

$$\phi = \frac{1}{3}x^3yz + \frac{1}{2}x^2z + h_1(y, z)$$

with $h_1(y, z)$ independent of $x$. Differentiating this with respect to $y$ and $z$ gives

$$\phi_y = \frac{1}{3}x^3z + \frac{\partial h_1}{\partial y}$$

$$\phi_z = \frac{1}{3}x^3y + \frac{1}{2}x^2 + \frac{\partial h_1}{\partial z}$$

from which we conclude $\dfrac{\partial h_1}{\partial y} = y + z$ and $\dfrac{\partial h_1}{\partial z} = y$.

Integrating with respect to $y$ and $z$ respectively gives

$$h_1 = \frac{1}{2}y^2 + yz + r(z) = yz + s(y)$$

with $r(z)$ independent of $x, y$ and $s(y)$ independent of $x, z$. Putting these together we get

$$h_1(y, z) = \frac{1}{2}y^2 + yz + C$$

where $C$ is a constant, and hence

$$\phi(x, y, z) = \frac{1}{3}x^3yz + \frac{1}{2}x^2z + \frac{1}{2}y^2 + yz + C.$$

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**End of Assignment — Total Available Marks = 45**